



US012400613B2

(12) **United States Patent**
Dunn et al.

(10) **Patent No.:** **US 12,400,613 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **DISPLAY UNITS WITH AUTOMATED
POWER GOVERNING**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **19/002,538**

(22) Filed: **Dec. 26, 2024**

(65) **Prior Publication Data**

US 2025/0149005 A1 May 8, 2025

Related U.S. Application Data

(63) Continuation of application No. 18/885,160, filed on
Sep. 13, 2024, now Pat. No. 12,217,713, which is a
continuation of application No. 18/652,287, filed on
May 1, 2024, now Pat. No. 12,118,953, which is a
continuation of application No. 18/214,948, filed on
Jun. 27, 2023, now Pat. No. 12,027,132.

(51) **Int. Cl.**
G09G 3/36 (2006.01)
G09G 1/00 (2006.01)

(52) **U.S. Cl.**
CPC **G09G 3/36** (2013.01); **G09G 1/005**
(2013.01); **G09G 2320/041** (2013.01); **G09G**
2330/021 (2013.01)

(58) **Field of Classification Search**

CPC G09G 2320/041
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,504,528 A	4/1970	Weinberg et al.
3,807,220 A	4/1974	Ottenstein et al.
4,093,355 A	6/1978	Kaplit et al.
4,527,804 A	7/1985	Spencer
4,593,978 A	6/1986	Mourey et al.
(Continued)		

FOREIGN PATENT DOCUMENTS

CN	203277867 U	11/2013
CN	112526806 A	3/2021
(Continued)		

OTHER PUBLICATIONS

Novitsky, T. et al., Design How-To, Driving LEDs versus CCFLs
for LCD backlighting, EE Times, Nov. 12, 2007, 6 pages, AspenCore.
(Continued)

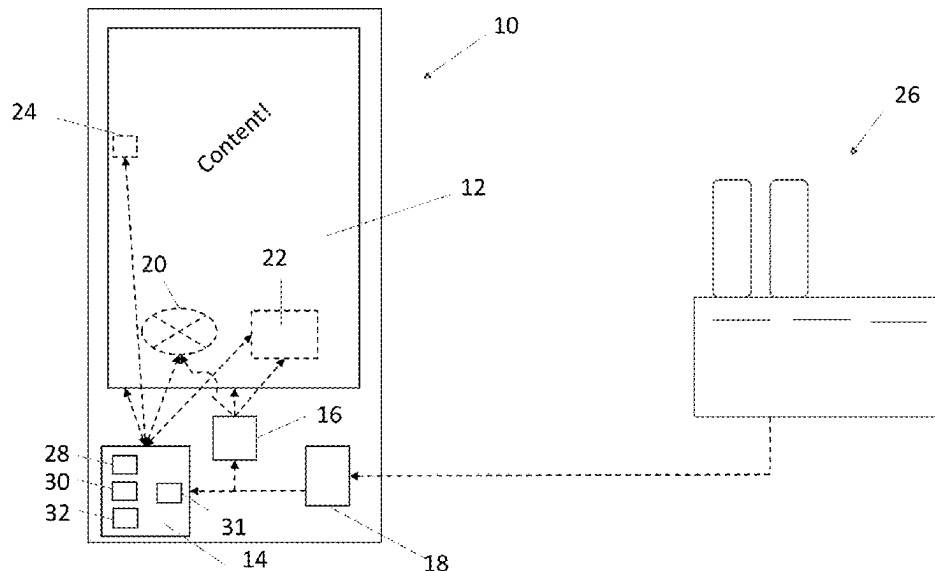
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(57) **ABSTRACT**

Display units for prioritizing requested operational updates
are disclosed. A control subsystem is in electronic commu-
nication with an electronic display, sensor(s), and a thermal
management subsystem. The control subsystem operates the
display unit based on operational parameters which are
specific to readings from the one or more sensors. Requests
to update operational parameters for the display unit are
received at the control subsystem and are prioritized for
implementation.

17 Claims, 4 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

4,634,225	A	1/1987	Haim et al.	7,015,470	B2	3/2006	Faytlin et al.
4,722,669	A	2/1988	Kundert	7,064,672	B2	6/2006	Gothard
5,029,982	A	7/1991	Nash	7,083,285	B2	8/2006	Hsu et al.
5,086,314	A	2/1992	Aoki et al.	7,136,076	B2	11/2006	Evanicky et al.
5,088,806	A	2/1992	McCartney et al.	7,307,614	B2	12/2007	Vinn
5,162,785	A	11/1992	Fagard	7,319,862	B1	1/2008	Lincoln et al.
5,168,961	A	12/1992	Schneider	7,324,080	B1	1/2008	Hu et al.
5,228,339	A	7/1993	Maresca, Jr. et al.	7,330,002	B2	2/2008	Joung
5,247,374	A	9/1993	Terada	7,354,159	B2	4/2008	Nakamura et al.
5,285,677	A	2/1994	Oehler	7,380,265	B2	5/2008	Jensen et al.
5,322,051	A	6/1994	Patterson et al.	7,391,317	B2	6/2008	Abraham et al.
5,351,201	A	9/1994	Harshbarger, Jr. et al.	7,447,018	B2	11/2008	Lee et al.
5,559,614	A	9/1996	Urbish et al.	7,451,332	B2	11/2008	Culbert et al.
5,590,831	A	1/1997	Manson et al.	7,474,294	B2	1/2009	Leo et al.
5,661,374	A	8/1997	Cassidy et al.	7,480,042	B1	1/2009	Phillips et al.
5,748,269	A	5/1998	Harris et al.	7,516,223	B2	4/2009	Whitehead
5,751,346	A	5/1998	Dozier et al.	7,518,600	B2	4/2009	Lee
5,767,489	A	6/1998	Ferrier	7,577,458	B2	8/2009	Lin
5,769,705	A	6/1998	O'Callaghan et al.	7,581,094	B1	8/2009	Apostolopoulos et al.
5,783,909	A	7/1998	Hochstein	7,595,785	B2	9/2009	Jang
5,786,801	A	7/1998	Ichise	7,612,278	B2	11/2009	Sitrick et al.
5,808,418	A	9/1998	Pitman et al.	7,636,927	B2	12/2009	Zigmond et al.
5,818,010	A	10/1998	McCann	7,639,220	B2	12/2009	Yoshida et al.
5,952,992	A	9/1999	Helms	7,658,787	B2	2/2010	Morse et al.
5,991,153	A	11/1999	Heady et al.	7,675,862	B2	3/2010	Pham et al.
6,042,443	A	3/2000	Carella et al.	7,679,279	B2	3/2010	Kamio et al.
6,085,152	A	7/2000	Doerfel	7,692,621	B2	4/2010	Song
6,089,751	A	7/2000	Conover et al.	7,751,813	B2	7/2010	Varanda
6,144,359	A	11/2000	Grave	7,764,280	B2	7/2010	Shiina
6,153,985	A	11/2000	Grossman	7,774,633	B1	8/2010	Harrenstien et al.
6,157,143	A	12/2000	Bigio et al.	7,795,574	B2	9/2010	Kennedy et al.
6,157,432	A	12/2000	Helbing	7,795,821	B2	9/2010	Jun
6,158,692	A	12/2000	Abild et al.	7,800,706	B2	9/2010	Kim et al.
6,181,070	B1	1/2001	Dunn et al.	7,804,477	B2	9/2010	Sawada et al.
6,191,839	B1	2/2001	Briley et al.	7,882,728	B2	2/2011	Kizaki et al.
6,215,411	B1	4/2001	Gothard	7,889,852	B2	2/2011	Whitehead
6,222,841	B1	4/2001	Taniguchi	7,949,893	B1	5/2011	Knaus et al.
6,259,492	B1	7/2001	Imoto et al.	8,074,627	B2	12/2011	Siddiqui et al.
6,292,228	B1	9/2001	Cho	8,111,371	B2	2/2012	Suminoe et al.
6,297,859	B1	10/2001	George	8,144,110	B2	3/2012	Huang
6,374,187	B1	4/2002	Knight et al.	8,175,841	B2	5/2012	Ooghe
6,380,853	B1	4/2002	Long et al.	8,212,921	B2	7/2012	Yun
6,384,736	B1	5/2002	Gothard	8,218,812	B2	7/2012	Sugimoto et al.
6,388,388	B1	5/2002	Weindorf et al.	8,248,203	B2	8/2012	Hanwright et al.
6,400,101	B1	6/2002	Biebl et al.	8,319,936	B2	11/2012	Yoshida et al.
6,417,900	B1	7/2002	Shin et al.	8,325,057	B2	12/2012	Salter
6,421,694	B1	7/2002	Nawaz et al.	8,336,369	B2	12/2012	Mahoney
6,496,236	B1	12/2002	Cole et al.	8,441,574	B2	5/2013	Dunn et al.
6,509,911	B1	1/2003	Shimotono	8,483,554	B2	7/2013	Takimoto et al.
6,526,807	B1	3/2003	Doumit et al.	8,508,155	B2	8/2013	Schuch
6,535,266	B1	3/2003	Nemeth et al.	8,601,252	B2	12/2013	Mendelow et al.
6,546,294	B1	4/2003	Kelsey et al.	8,612,608	B2	12/2013	Whitehead
6,553,336	B1	4/2003	Johnson et al.	8,643,589	B2	2/2014	Wang
6,556,258	B1	4/2003	Yoshida et al.	8,654,302	B2	2/2014	Dunn et al.
6,587,525	B2	7/2003	Jeong et al.	8,689,343	B2	4/2014	De Laet
6,628,355	B1	9/2003	Takahara	8,700,226	B2	4/2014	Schuch et al.
6,701,143	B1	3/2004	Dukach et al.	8,767,165	B2	7/2014	Dunn
6,712,046	B2	3/2004	Nakamichi	8,810,501	B2	8/2014	Budzelaar et al.
6,753,842	B1	6/2004	Williams et al.	8,823,630	B2	9/2014	Roberts et al.
6,762,741	B2	7/2004	Weindorf	8,854,595	B2	10/2014	Dunn
6,771,795	B1	8/2004	Isnardi	8,881,576	B2	11/2014	Schwartz et al.
6,809,718	B2	10/2004	Wei et al.	8,901,825	B2	12/2014	Reed
6,812,851	B1	11/2004	Dukach et al.	8,988,011	B2	3/2015	Dunn
6,813,375	B2	11/2004	Armato III et al.	9,026,686	B2	5/2015	Dunn et al.
6,821,179	B2	11/2004	Ando	9,147,194	B1	9/2015	Le et al.
6,839,104	B2	1/2005	Taniguchi et al.	9,363,262	B1	6/2016	Wilkes
6,850,209	B2	2/2005	Mankins et al.	9,400,192	B1	7/2016	Salser, Jr. et al.
6,886,942	B2	5/2005	Okada et al.	9,445,470	B2	9/2016	Wang et al.
6,891,135	B2	5/2005	Pala et al.	9,448,569	B2	9/2016	Schuch et al.
6,943,768	B2	9/2005	Cavanaugh et al.	9,516,485	B1	12/2016	Bowers et al.
6,955,170	B1	10/2005	Mullins et al.	9,622,392	B1	4/2017	Bowers et al.
6,968,375	B1	11/2005	Brown	9,629,287	B2	4/2017	Dunn
6,982,686	B2	1/2006	Miyachi et al.	9,760,151	B1	9/2017	Hou
6,996,460	B1	2/2006	Krahnstoever et al.	9,881,528	B2	1/2018	Dunn
7,007,545	B1	3/2006	Martinek	9,924,583	B2	3/2018	Schuch et al.
				10,170,076	B2	1/2019	Wang et al.
				10,174,519	B1	1/2019	Carpenter et al.
				10,194,562	B2	1/2019	Shelnutt et al.
				10,255,884	B2	4/2019	Dunn et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,311,763	B2	6/2019	Greenfield	2005/0024538	A1	2/2005	Park et al.
10,321,549	B2	6/2019	Schuch et al.	2005/0033840	A1	2/2005	Nisani et al.
10,409,544	B2	9/2019	Park et al.	2005/0043907	A1	2/2005	Eckel et al.
10,412,816	B2	9/2019	Schuch et al.	2005/0049729	A1	3/2005	Culbert et al.
10,440,790	B2	10/2019	Dunn et al.	2005/0070335	A1	3/2005	Jitsuishi et al.
10,578,658	B2	3/2020	Dunn et al.	2005/0071252	A1	3/2005	Henning et al.
10,593,175	B1	3/2020	Jennings et al.	2005/0073518	A1	4/2005	Bontempi
10,607,520	B2	3/2020	Schuch et al.	2005/0088984	A1	4/2005	Chin et al.
10,795,413	B1	10/2020	Dunn	2005/0094391	A1	5/2005	Campbell et al.
10,803,783	B2	10/2020	Wang et al.	2005/0123001	A1	6/2005	Craven et al.
10,858,886	B2	12/2020	Fasi et al.	2005/0127796	A1	6/2005	Olesen et al.
10,860,141	B2	12/2020	Wang et al.	2005/0132036	A1	6/2005	Jang et al.
11,016,547	B2	5/2021	Whitehead et al.	2005/0140640	A1	6/2005	Oh et al.
11,131,453	B2	9/2021	Kim et al.	2005/0179554	A1	8/2005	Lu
11,132,715	B2	9/2021	Menendez et al.	2005/0184983	A1	8/2005	Brabander et al.
11,402,940	B2	8/2022	Dunn	2005/0216939	A1	9/2005	Corbin
11,645,029	B2	5/2023	Newnham et al.	2005/0231457	A1	10/2005	Yamamoto et al.
11,803,344	B2	10/2023	Newnham et al.	2005/0242741	A1	11/2005	Shiota et al.
11,965,804	B2	4/2024	Dunn et al.	2005/0267943	A1	12/2005	Castaldi et al.
11,972,672	B1	4/2024	Dunn	2005/0289061	A1	12/2005	Kulakowski et al.
12,027,132	B1	7/2024	Dunn et al.	2005/0289588	A1	12/2005	Kinnear
2002/0009978	A1	1/2002	Dukach et al.	2006/0007107	A1	1/2006	Ferguson
2002/0019933	A1	2/2002	Friedman et al.	2006/0022616	A1	2/2006	Furukawa et al.
2002/0020090	A1	2/2002	Sanders	2006/0038511	A1	2/2006	Tagawa
2002/0026354	A1	2/2002	Shoji et al.	2006/0049533	A1	3/2006	Kamoshita
2002/0050974	A1	5/2002	Rai et al.	2006/0087521	A1	4/2006	Chu et al.
2002/0065046	A1	5/2002	Mankins et al.	2006/0125773	A1	6/2006	Ichikawa et al.
2002/0084891	A1	7/2002	Mankins et al.	2006/0130501	A1	6/2006	Singh et al.
2002/0101553	A1	8/2002	Enomoto et al.	2006/0150222	A1	7/2006	McCafferty et al.
2002/0112026	A1	8/2002	Fridman et al.	2006/0160614	A1	7/2006	Walker et al.
2002/0120721	A1	8/2002	Eilers et al.	2006/0197474	A1	9/2006	Olsen
2002/0126248	A1	9/2002	Yoshida	2006/0197735	A1	9/2006	Vuong et al.
2002/0147648	A1	10/2002	Fadden et al.	2006/0207730	A1	9/2006	Berman et al.
2002/0152425	A1	10/2002	Chaiken et al.	2006/0214904	A1	9/2006	Kimura et al.
2002/0154138	A1	10/2002	Wada et al.	2006/0215044	A1	9/2006	Masuda et al.
2002/0163513	A1	11/2002	Tsuji	2006/0220571	A1	10/2006	Howell et al.
2002/0163916	A1	11/2002	Oskouy et al.	2006/0238531	A1	10/2006	Wang
2002/0164962	A1	11/2002	Mankins et al.	2006/0244702	A1	11/2006	Yamazaki et al.
2002/0167637	A1	11/2002	Burke et al.	2006/0269216	A1	11/2006	Wiemeyer et al.
2002/0190972	A1	12/2002	Ven de Van	2007/0013828	A1	1/2007	Cho et al.
2002/0194365	A1	12/2002	Jammes	2007/0039028	A1	2/2007	Bar
2002/0194609	A1	12/2002	Tran	2007/0047808	A1	3/2007	Choe et al.
2003/0007109	A1	1/2003	Park	2007/0152949	A1	7/2007	Sakai
2003/0031128	A1	2/2003	Kim et al.	2007/0153117	A1	7/2007	Lin et al.
2003/0039312	A1	2/2003	Horowitz et al.	2007/0154060	A1	7/2007	Sun
2003/0061316	A1	3/2003	Blair et al.	2007/0157260	A1	7/2007	Walker
2003/0088832	A1	5/2003	Agostinelli et al.	2007/0168539	A1	7/2007	Day
2003/0097497	A1	5/2003	Esakov	2007/0171647	A1	7/2007	Artwohl et al.
2003/0098881	A1	5/2003	Nolte et al.	2007/0173297	A1	7/2007	Cho et al.
2003/0115591	A1	6/2003	Weissmueller, Jr. et al.	2007/0200513	A1	8/2007	Ha et al.
2003/0117714	A1	6/2003	Nakamura et al.	2007/0214812	A1	9/2007	Wagner et al.
2003/0122810	A1	7/2003	Tsirkel et al.	2007/0222730	A1	9/2007	Kao et al.
2003/0132514	A1	7/2003	Liebeskind	2007/0230167	A1	10/2007	McMahon et al.
2003/0161354	A1	8/2003	Bader et al.	2007/0237636	A1	10/2007	Hsu
2003/0177269	A1	9/2003	Robinson et al.	2007/0242153	A1	10/2007	Tang et al.
2003/0192060	A1	10/2003	Levy	2007/0247594	A1	10/2007	Tanaka
2003/0196208	A1	10/2003	Jacobson	2007/0268234	A1	11/2007	Wakabayashi et al.
2003/0204342	A1	10/2003	Law et al.	2007/0268241	A1	11/2007	Nitta et al.
2003/0214242	A1	11/2003	Berg-johansen	2007/0273519	A1	11/2007	Ichikawa et al.
2003/0230991	A1	12/2003	Muthu et al.	2007/0273624	A1	11/2007	Geelen
2004/0032382	A1	2/2004	Cok et al.	2007/0274400	A1	11/2007	Murai et al.
2004/0036622	A1	2/2004	Dukach et al.	2007/0279369	A1	12/2007	Yao et al.
2004/0036697	A1	2/2004	Kim et al.	2007/0286107	A1	12/2007	Singh et al.
2004/0036834	A1	2/2004	Ohnishi et al.	2007/0291198	A1	12/2007	Shen
2004/0113044	A1	6/2004	Ishiguchi	2007/0297163	A1	12/2007	Kim et al.
2004/0138840	A1	7/2004	Wolfe	2007/0297172	A1	12/2007	Furukawa et al.
2004/0158872	A1	8/2004	Kobayashi	2008/0008471	A1	1/2008	Dress
2004/0165139	A1	8/2004	Anderson et al.	2008/0019147	A1	1/2008	Erchak et al.
2004/0194131	A1	9/2004	Ellis et al.	2008/0024268	A1	1/2008	Wong et al.
2004/0201547	A1	10/2004	Takayama	2008/0034205	A1	2/2008	Alain et al.
2004/0210419	A1	10/2004	Wiebe et al.	2008/0037466	A1	2/2008	Ngo et al.
2004/0243940	A1	12/2004	Lee et al.	2008/0037783	A1	2/2008	Kim et al.
2004/0252400	A1	12/2004	Blank et al.	2008/0055297	A1	3/2008	Park
2004/0253947	A1	12/2004	Phillips et al.	2008/0074382	A1	3/2008	Lee et al.
2005/0012734	A1	1/2005	Johnson et al.	2008/0078921	A1	4/2008	Yang et al.
				2008/0084166	A1	4/2008	Tsai
				2008/0104631	A1	5/2008	Krock et al.
				2008/0111958	A1	5/2008	Kleverman et al.
				2008/0112601	A1	5/2008	Warp

(56)

References Cited

U.S. PATENT DOCUMENTS

2008/0136770 A1	6/2008	Peker et al.	2011/0163691 A1	7/2011	Dunn
2008/0143187 A1	6/2008	Hoekstra et al.	2011/0175872 A1	7/2011	Chuang et al.
2008/0151082 A1	6/2008	Chan	2011/0193872 A1	8/2011	Biernath et al.
2008/0163291 A1	7/2008	Fishman et al.	2011/0231676 A1	9/2011	Atkins et al.
2008/0165203 A1	7/2008	Pantfoerder	2011/0260534 A1	10/2011	Rozman et al.
2008/0170031 A1	7/2008	Kuo	2011/0264273 A1	10/2011	Grabinger et al.
2008/0176345 A1	7/2008	Yu et al.	2011/0279426 A1	11/2011	Imamura et al.
2008/0185976 A1	8/2008	Dickey et al.	2011/0283199 A1	11/2011	Schuch et al.
2008/0204375 A1	8/2008	Shin et al.	2012/0075362 A1	3/2012	Ichioaka et al.
2008/0218501 A1	9/2008	Diamond	2012/0081279 A1	4/2012	Greenebaum et al.
2008/0224892 A1	9/2008	Bogolea et al.	2012/0105424 A1	5/2012	Lee et al.
2008/0230497 A1	9/2008	Strickland et al.	2012/0162204 A1	6/2012	Vesely et al.
2008/0246871 A1	10/2008	Kupper et al.	2012/0176420 A1	7/2012	Liu
2008/0259198 A1	10/2008	Chen et al.	2012/0182278 A1	7/2012	Ballestad
2008/0266554 A1	10/2008	Sekine et al.	2012/0197459 A1	8/2012	Fukano
2008/0267328 A1	10/2008	Ianni et al.	2012/0203872 A1	8/2012	Luby et al.
2008/0278099 A1	11/2008	Bergfors et al.	2012/0211001 A1	8/2012	Elshafei
2008/0278100 A1	11/2008	Hwang	2012/0212520 A1	8/2012	Matsui et al.
2008/0281165 A1	11/2008	Rai et al.	2012/0252495 A1	10/2012	Moeglein et al.
2008/0303918 A1	12/2008	Keithley	2012/0268436 A1	10/2012	Chang
2008/0313691 A1	12/2008	Cholas et al.	2012/0269382 A1	10/2012	Kiyohara et al.
2009/0009997 A1	1/2009	Sanfilippo et al.	2012/0284547 A1	11/2012	Culbert et al.
2009/0014548 A1	1/2009	Criss et al.	2012/0302343 A1	11/2012	Hurst et al.
2009/0015400 A1	1/2009	Breed	2012/0308191 A1	12/2012	Chung et al.
2009/0033612 A1	2/2009	Roberts et al.	2013/0007110 A1	1/2013	Centner
2009/0034283 A1	2/2009	Albright et al.	2013/0027370 A1	1/2013	Dunn et al.
2009/0036190 A1	2/2009	Brosnan et al.	2013/0070567 A1	3/2013	Marzouq
2009/0079416 A1	3/2009	Vinden et al.	2013/0098425 A1	4/2013	Amin et al.
2009/0085859 A1	4/2009	Song	2013/0113973 A1	5/2013	Miao
2009/0091634 A1	4/2009	Kennedy et al.	2013/0158730 A1	6/2013	Yasuda et al.
2009/0104989 A1	4/2009	Williams et al.	2013/0162908 A1	6/2013	Son et al.
2009/0109129 A1	4/2009	Cheong et al.	2013/0173358 A1	7/2013	Pinkus
2009/0129556 A1	5/2009	Ahn	2013/0278868 A1	10/2013	Dunn et al.
2009/0135167 A1	5/2009	Sakai et al.	2013/0279090 A1	10/2013	Brandt
2009/0152445 A1	6/2009	Gardner, Jr.	2013/0282154 A1	10/2013	Chappell et al.
2009/0164615 A1	6/2009	Akkanen	2013/0344794 A1	12/2013	Shaw et al.
2009/0273568 A1	11/2009	Milner	2014/0002747 A1	1/2014	Macholz
2009/0278766 A1	11/2009	Sako et al.	2014/0009893 A1	1/2014	Lai
2009/0284457 A1	11/2009	Botzas et al.	2014/0132796 A1	5/2014	Prentice et al.
2009/0289968 A1	11/2009	Yoshida	2014/0139116 A1	5/2014	Reed
2009/0315867 A1	12/2009	Sakamoto et al.	2014/0172174 A1	6/2014	Poss et al.
2010/0017526 A1	1/2010	Jagannath et al.	2014/0184980 A1	7/2014	Onoue
2010/0033413 A1	2/2010	Song et al.	2014/0190240 A1	7/2014	He et al.
2010/0037274 A1	2/2010	Meuninck et al.	2014/0204452 A1	7/2014	Branson
2010/0039366 A1	2/2010	Hardy	2014/0230526 A1	8/2014	Willemin et al.
2010/0039414 A1	2/2010	Bell	2014/0232709 A1	8/2014	Dunn et al.
2010/0039440 A1	2/2010	Tanaka et al.	2014/0293605 A1	10/2014	Chemel et al.
2010/0060550 A1	3/2010	McGinn et al.	2014/0365965 A1	12/2014	Bray et al.
2010/0060861 A1	3/2010	Medin	2015/0009627 A1	1/2015	Dunn et al.
2010/0066484 A1	3/2010	Hanwright et al.	2015/0062892 A1	3/2015	Krames et al.
2010/0083305 A1	4/2010	Acharya et al.	2015/0070337 A1	3/2015	Bell et al.
2010/0149567 A1	6/2010	Kanazawa et al.	2015/0169827 A1	6/2015	LaBorde
2010/0177157 A1	7/2010	Stephens et al.	2015/0193074 A1	7/2015	Cudak et al.
2010/0177158 A1	7/2010	Walter	2015/0237761 A1	8/2015	Dunn
2010/0177750 A1	7/2010	Essinger et al.	2015/0250021 A1	9/2015	Stice et al.
2010/0194725 A1	8/2010	Yoshida et al.	2015/0310313 A1	10/2015	Murayama et al.
2010/0198983 A1	8/2010	Monroe et al.	2015/0319882 A1	11/2015	Dunn et al.
2010/0226091 A1	9/2010	Dunn	2015/0346525 A1	12/2015	Wolf et al.
2010/0231563 A1	9/2010	Dunn et al.	2015/0348460 A1	12/2015	Cox et al.
2010/0231602 A1	9/2010	Huang	2016/0034240 A1	2/2016	Kreiner et al.
2010/0237697 A1	9/2010	Dunn et al.	2016/0037606 A1	2/2016	Dunn et al.
2010/0253660 A1	10/2010	Hashimoto	2016/0055671 A1	2/2016	Menozi et al.
2010/0299556 A1	11/2010	Taylor et al.	2016/0112521 A1	4/2016	Lawson et al.
2010/0309361 A1	12/2010	Fukushima	2016/0125468 A1	5/2016	Staneluis et al.
2011/0016006 A1	1/2011	Opdycke	2016/0125772 A1	5/2016	Li et al.
2011/0019636 A1	1/2011	Fukuoka et al.	2016/0162297 A1	6/2016	Shao
2011/0032489 A1	2/2011	Kimoto et al.	2016/0292744 A1	10/2016	Strimaitis et al.
2011/0047567 A1	2/2011	Zigmond et al.	2016/0293142 A1	10/2016	Bowden et al.
2011/0050738 A1	3/2011	Fujioka et al.	2016/0334811 A1	11/2016	Marten
2011/0058326 A1	3/2011	Idems et al.	2016/0335698 A1	11/2016	Jones et al.
2011/0074737 A1	3/2011	Hsieh et al.	2016/0338182 A1	11/2016	Schuch et al.
2011/0074803 A1	3/2011	Kerofsky	2016/0358530 A1	12/2016	Schuch et al.
2011/0078536 A1	3/2011	Han et al.	2016/0358538 A1	12/2016	Schuch et al.
2011/0102630 A1	5/2011	Rukes	2016/0360167 A1	12/2016	Mitchell et al.
2011/0148904 A1	6/2011	Kotani	2017/0060369 A1	3/2017	Goyal et al.
			2017/0075777 A1	3/2017	Dunn et al.
			2017/0083043 A1	3/2017	Bowers et al.
			2017/0091822 A1	3/2017	Tian et al.
			2017/0111486 A1	4/2017	Bowers et al.

(56)

References Cited**U.S. PATENT DOCUMENTS**

2017/0111520 A1 4/2017 Bowers et al.
 2017/0163519 A1 6/2017 Bowers et al.
 2017/0168295 A1 6/2017 Iwami
 2017/0242502 A1 8/2017 Gray et al.
 2017/0242534 A1 8/2017 Gray
 2017/0256051 A1 9/2017 Dwivedi et al.
 2017/0315886 A1 11/2017 Helmick et al.
 2018/0012565 A1 1/2018 Dunn
 2018/0027635 A1 1/2018 Roquemore, III
 2018/0042134 A1 2/2018 Dunn et al.
 2018/0088368 A1 3/2018 Notoshi et al.
 2018/0089717 A1 3/2018 Morin et al.
 2018/0128708 A1 5/2018 Cirino
 2018/0129461 A1 5/2018 Kim-Whitty
 2018/0130385 A1 5/2018 Qian et al.
 2018/0181091 A1 6/2018 Funk et al.
 2018/0203475 A1 7/2018 Van Derven et al.
 2018/0268783 A1 9/2018 Woo
 2018/0284758 A1 10/2018 Cella et al.
 2018/0314103 A1 11/2018 Dunn et al.
 2019/0018506 A1 1/2019 Bernstein et al.
 2019/0021189 A1 1/2019 Kim et al.
 2019/0087042 A1 3/2019 Van Ostrand et al.
 2019/0096202 A1 3/2019 Seelman
 2019/0116719 A1 4/2019 Fletcher et al.
 2019/0171331 A1 6/2019 Gray et al.
 2019/0289754 A1 9/2019 Hubbard
 2019/0339312 A1 11/2019 Dunn et al.
 2019/0367148 A1 12/2019 Kehlenbeck et al.
 2019/0383778 A1 12/2019 Dunn et al.
 2020/0012116 A1 1/2020 Fuerst et al.
 2020/0012383 A1 1/2020 Wang et al.
 2020/0019363 A1 1/2020 Newnham et al.
 2020/0033017 A1 1/2020 Brown et al.
 2020/0128701 A1 4/2020 Whitehead et al.
 2020/0272269 A1 8/2020 Dunn
 2020/0294401 A1 9/2020 Kerecsen
 2020/0390009 A1 12/2020 Whitehead et al.
 2021/0034101 A1 2/2021 Yildiz et al.
 2021/0035494 A1 2/2021 Yildiz et al.
 2021/0174715 A1 6/2021 Holloway et al.
 2021/0302779 A1 9/2021 Dunn
 2021/0397292 A1 12/2021 Dunn
 2022/0121255 A1 4/2022 Wang et al.
 2022/0260872 A1 8/2022 Dunn et al.
 2022/0295666 A1 9/2022 Dunn et al.
 2023/0029615 A1 2/2023 Dunn et al.
 2023/0048815 A1 2/2023 Newnham et al.
 2023/0052966 A1 2/2023 Newnham et al.
 2023/0160774 A1 5/2023 Dunn et al.
 2023/0296278 A1 9/2023 Sawada et al.
 2023/0333423 A1 10/2023 Dunn et al.
 2024/0144806 A1 5/2024 Dunn
 2024/0201040 A1 6/2024 Dunn et al.

FOREIGN PATENT DOCUMENTS

CN 217384567 U 9/2022
 EP 0313331 B1 2/1994
 EP 1686777 A1 8/2006
 EP 1821538 A1 8/2007
 EP 2299723 A1 3/2011
 EP 2351369 A2 8/2011
 GB 2369730 A 5/2002
 JP 61-234690 A 10/1986
 JP 61-251901 A 11/1986
 JP 3-153212 A 7/1991
 JP 5-18767 A 1/1993
 JP 7-74224 A 3/1995
 JP 8193727 A 7/1996
 JP 8-338981 A 12/1996
 JP 11-160727 A 6/1999
 JP 2000122575 A 4/2000
 JP 3080628 B2 8/2000
 JP 2002064842 A 2/2002

JP 2002209230 A 7/2002
 JP 2005-148490 A 6/2005
 JP 2005-211449 A 8/2005
 JP 2005-211451 A 8/2005
 JP 2005236469 A 9/2005
 JP 2005265922 A 9/2005
 JP 2005-338266 A 12/2005
 JP 2005333568 A 12/2005
 JP 2006-106345 A 4/2006
 JP 2006-145890 A 6/2006
 JP 2006318733 A 11/2006
 JP 2007003638 A 1/2007
 JP 2007322718 A 12/2007
 JP 2008-34841 A 2/2008
 JP 2008122695 A 5/2008
 JP 2009031622 A 2/2009
 JP 2010282109 A 12/2010
 JP 2014-149485 A 8/2014
 KR 200361111 Y1 9/2004
 KR 10-2006-0016469 A 2/2006
 KR 10-0768584 B1 10/2007
 KR 10-2008-0013592 A 2/2008
 KR 10-2008-0086245 A 9/2008
 KR 10-2009-0014903 A 2/2009
 KR 10-2010-0081354 A 7/2010
 KR 10-2011-0065338 6/2011
 KR 10-2014-0054747 A 5/2014
 WO WO9608892 A1 3/1996
 WO WO2008050402 A1 5/2008
 WO 2010/141739 A2 12/2010
 WO 2011/052331 A1 5/2011
 WO WO2012/127971 A1 9/2012
 WO WO2013/182733 A1 12/2013
 WO 2016/183576 A1 11/2016
 WO 2017/031237 A1 2/2017
 WO 2017/210317 A1 12/2017
 WO 2020/081687 A1 4/2020
 WO WO2023/009477 A1 2/2023

OTHER PUBLICATIONS

Vogler, A. et al., Photochemistry and Beer, Journal of Chemical Education, Jan. 1982, pp. 25-27, vol. 59, No. 1.
 Zeeff, T.M. et al., Abstract of EMC analysis of 18" LCD Monitor, Electromagnetic Compatibility, IEEE International Symposium, Aug. 21-25, 2000, vol. 1, 1 page.
 Lee, X., What is Gamma Correction in Images and Videos?, http://xahlee.info/img/what_is_gamma_correction.html, Feb. 24, 2010, 4 pages.
 Hooper, S. et al., Designing Mobile Interfaces, 2012, pp. 519-521, O'Reilly Media.
 Rouaissia, C., Adding Proximity Detection to a Standard Analog-Resistive Touchscreen, SID 2012 Digest, 2012, 1564-1566, p. 132.
 Photo Research, Inc., PR®-650 SpectraScan® Colorimeter, 1999, 2 pages.
 Texas Advanced Optoelectronic Solutions Inc., TCS230 Programmable Color Light-To-Frequency Converter, Dec. 2007, 12 pages.
 Methven, Don, Wireless Video Streaming: An Overview, Nov. 16, 2022, 7 pages.
 Outdoorlink, Inc., SmartLink One, One Relay, <http://smartlinkcontrol.com/billboard/one-relay/>, retrieved Apr. 17, 2019, 2007-16, 6 pages.
 Outdoorlink, Inc., SmartLink Website User Manual, <http://smartlink.outdoorlinkinc.com/docs/SmartLinkWebsiteUserManual.pdf>, 2017, 33 pages.
 Outdoorlink, Inc., SmartLink One Out of Home Media Controller, 2016, 1 page.
 Sigmasense, Analog can't touch Digital, <https://sigmasense.com/>, retrieved Jan. 23, 2019, 5 pages.
 Sigmasense, Solutions, <https://sigmasense.com/solutions/>, retrieved Jan. 23, 2019, 4 pages.
 Sigmasense, Technology, <https://sigmasense.com/technology/>, retrieved Jan. 23, 2019, 3 pages.

(56)

References Cited

OTHER PUBLICATIONS

Turley, Jim, SigmaSense ICCI Goes Big, New Touch Technology Aimed at Big Screens, But That's Just for Starters, EEJournal, <https://www.eejournal.com/article/sigma-sense-icci-goes-big/>, Jan. 8, 2019, 3 pages.

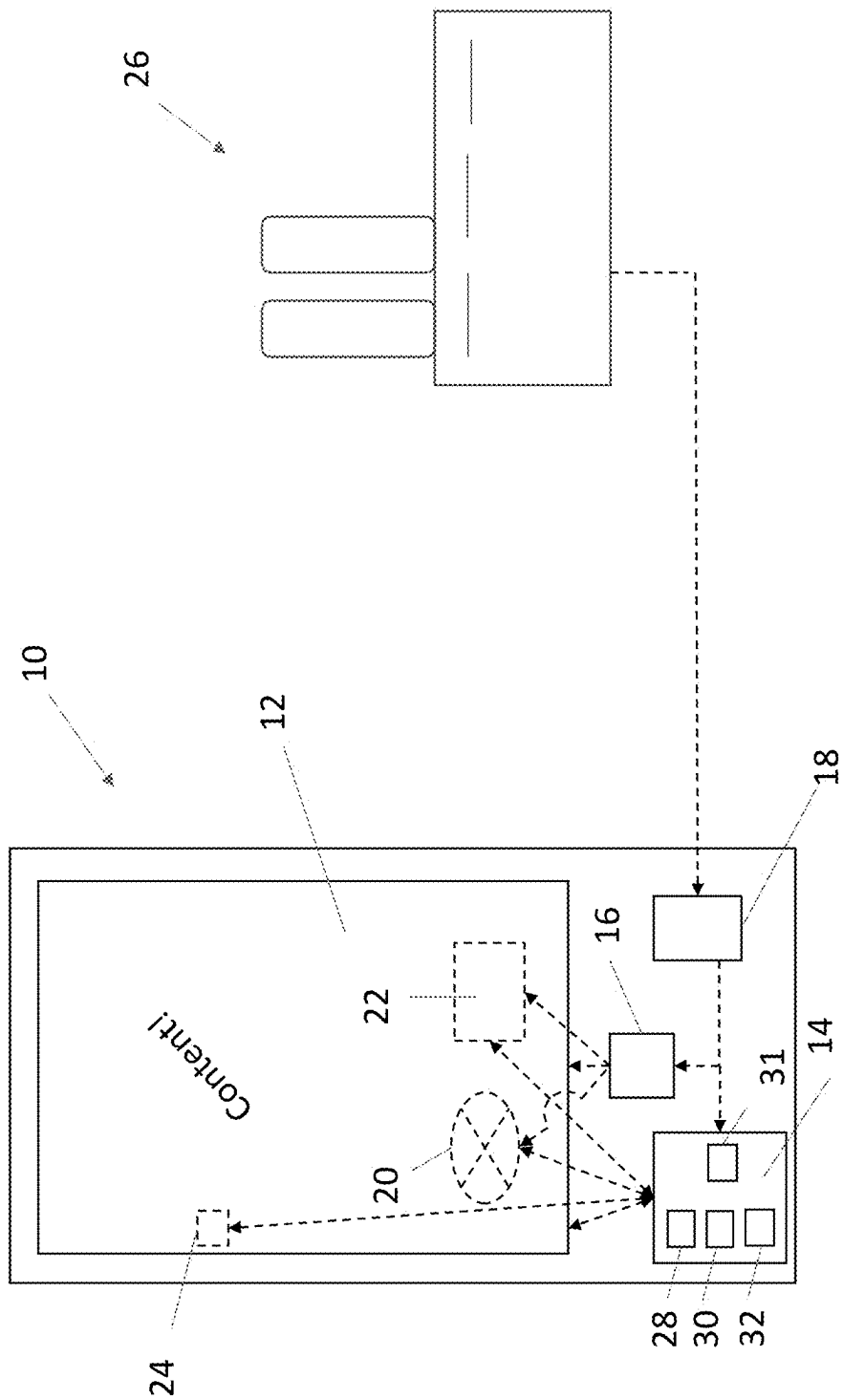


Figure 1

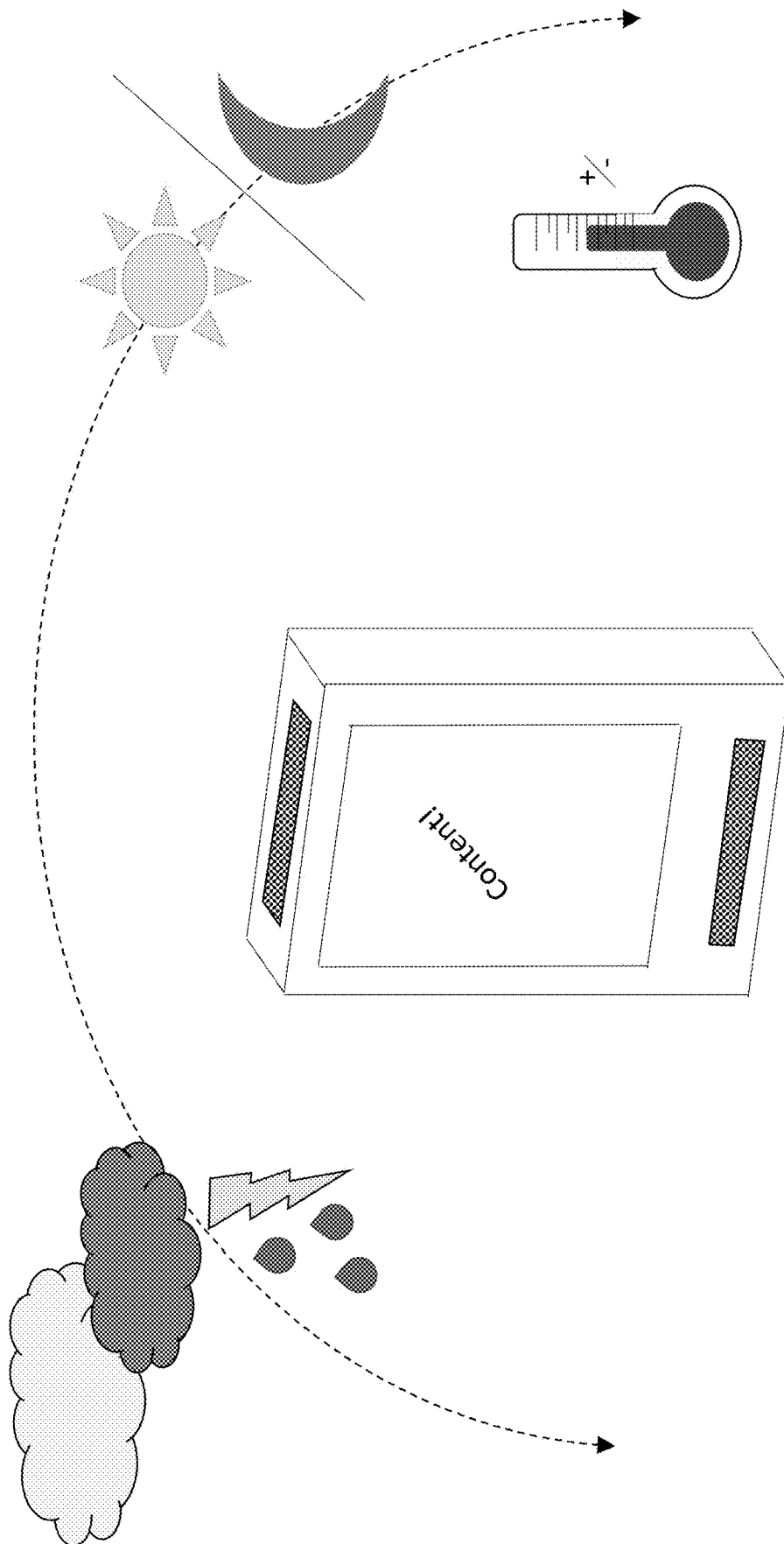


Figure 2

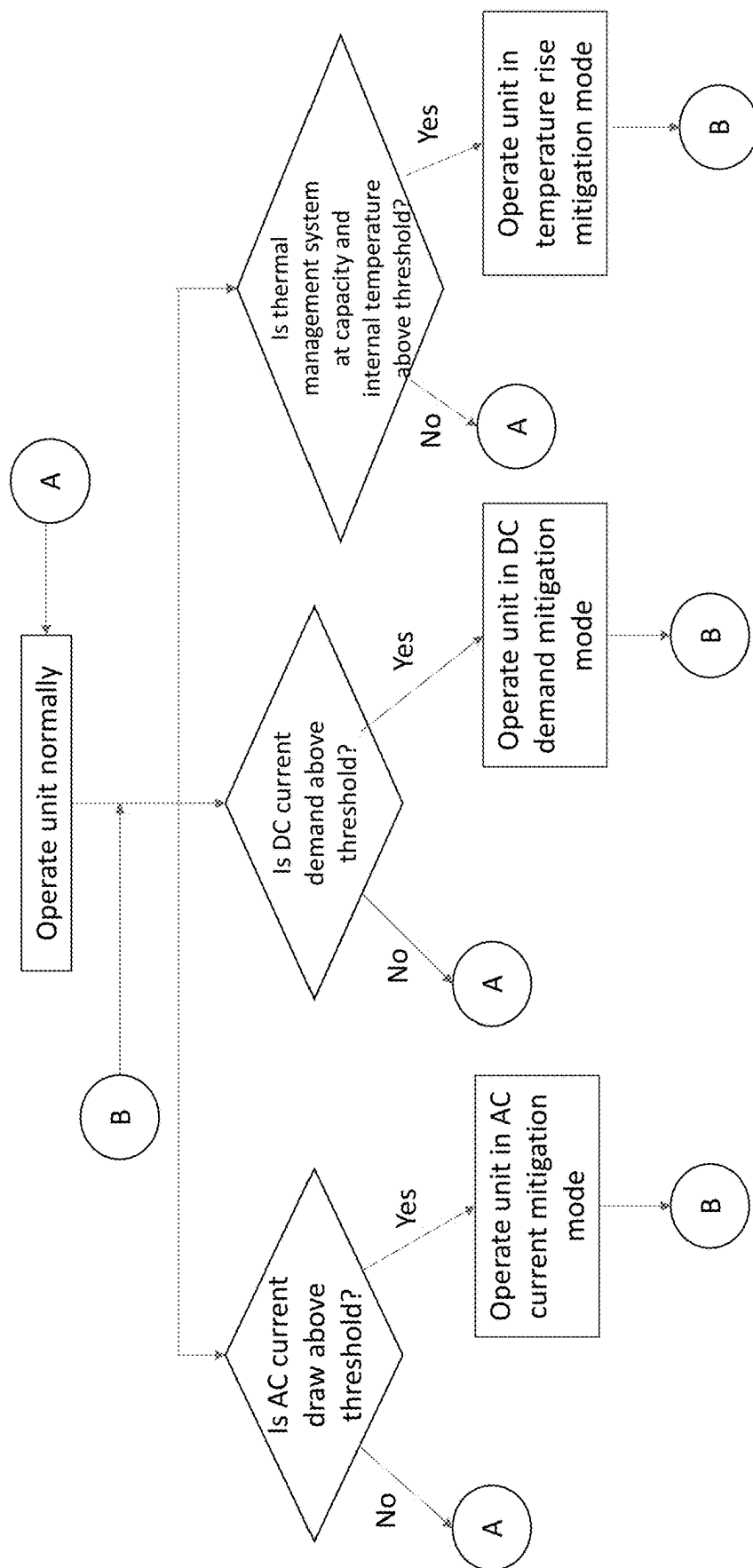


Figure 3

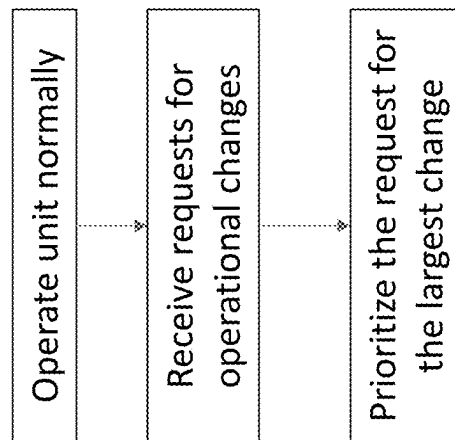


Figure 4

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DISPLAY UNITS WITH AUTOMATED POWER GOVERNING

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/885,160 filed Sep. 13, 2024, which is a continuation of U.S. application Ser. No. 18/652,287 filed May 1, 2024, now U.S. Pat. No. 12,118,953 issued Oct. 15, 2024, which is a continuation of U.S. application Ser. No. 18/214,948 filed Jun. 27, 2023, now U.S. Pat. No. 12,027,132 issued Jul. 2, 2024, the disclosures of which are hereby incorporated by reference as if fully restated herein.

TECHNICAL FIELD

Exemplary embodiments relate generally to display units with automated power governing as well as systems and methods related to the same.

BACKGROUND AND SUMMARY OF THE INVENTION

Digital out of home advertising has grown in recent years, and continues to be a major source of interest, not only for advertising, but also for other public announcements, various marketing and other services, “smart city” services, telecommunication services, and the like. These units are exposed to a wide variety of operating conditions (e.g., weather, user, and/or environment based) and demands (e.g., usage and/or programmed operational parameters). It is known to provide electronic displays in ruggedized enclosures, such as with thermal management and/or remote monitoring and/or control functions, to provide survivability and adaptability of such units under such demanding conditions. For example, it is known to limit alternating current (AC) draw on various electronic devices. It is also known to limit direct current (DC) draw on various electronic devices. It is also known to set thermal limits on various electronic devices where conditions are changed in response to meeting the thermal limit. However, current solutions fail to gracefully control such units in a fashion which exerts sufficient automated control to prevent or limit disruptions to user experiences while also safeguarding units from failure. What is needed is a power governing control system which reliably and operably controls such units to reduce or prevent failure while also minimizing disruption to user experience.

Units with power governing control systems which reliably control such units to reduce or prevent failure while also minimizing disruption to user experience are provided, along with systems and methods related to the same. Units may include one or more electronic displays. The units may include a control system with one or more of: an AC governor, a DC governor, and a thermal governor. The control system may be electrically interposed between a power source (e.g., external utility power supply) for the unit and some or all electricity consuming components of the unit (e.g., electronic displays, thermal management systems, customer equipment, peripheral equipment, combinations thereof, or the like). The governors may operably control illumination sources for the electronic displays, such as the backlights.

For the AC governor, where an AC current threshold is met or exceeded, the AC governor may operate in an AC current mitigation mode, such as by reducing power sup-

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plied to the illumination sources and/or other electricity consuming components rapidly, such as in a matter of seconds, and/or less than one second. The AC current threshold may be set relative to service rated current, such as between 70%-99% of the service rating. Power levels may be automatically increased where the AC current threshold is no longer met.

One or more circuit breakers or the like may be electrically interposed between the power source for the unit and some or all electricity consuming components of the unit to serve as a backup in case of continued AC current increase. The AC current threshold may be set to below the circuit breaker ratings, such as at 70%-99% thereof. AC current input may be monitored by the control system and electronic notifications may be generated and/or transmitted to remote device(s) where the AC current input is below a predetermined threshold, such as an expected current input or a margin thereof.

For the DC governor, where a DC current demand threshold is met or exceeded, the DC governor may operate in a DC current mitigation mode such as by reducing, at a relatively moderate rate, such as in a matter of multiple seconds or minutes, power to the illumination sources and/or other electricity consuming components, such as down to a zero level. Power levels may be automatically increased where the DC current threshold is no longer met.

Where one or more service limits of DC power supplies are met or exceeded, which may be a threshold above the DC current demand threshold, the DC power supplies may be automatically shut off and begin a restart sequence.

For the thermal governor, where the thermal management system of a unit is at maximum capacity or some other threshold capacity (e.g., 70%-99% thereof), and any one or more internal temperatures, such as measuring by one or more temperature sensors in electronic communication with the control system, the thermal governor may operate in a thermal mitigation mode, such as by reducing to relatively slow, such as over a matter of several minutes, reduce power provided to the illumination sources, such as down to a zero level. Power levels may be automatically increased where the DC current threshold is no longer met, such as over a period of a same or different number of minutes.

The power adjustments shown and/or described herein, such as by the AC, DC, and/or thermal governors may be provided on various bases, such as but not limited to, on a linear, exponential and/or the like basis relative to the respective threshold(s). A grace value may be set such that the governors are configured to remain within the various mitigation modes unless/until the relevant values pass the respective thresholds by at least a predetermined amount. This may reduce or prevent rapid and frequent transitions between the normal mode and mitigation mode(s).

The governors may be operated independently and may independently provide benefits for reducing or eliminating failures and interruptions to user experiences. Analysis of measures against thresholds for each governor may be performed in parallel or in any sequence. The combination and operation of the governors, in particular, may provide exceptional reduction or elimination of failures and interruptions to user experiences. The integrated safety feature may alternatively, or additionally, permit the reduction in power supplies for a unit and/or reduce power consumption for a unit, among other benefits. This may allow provided power supplies to operate closer to capacity, whereby increased efficiencies are generally found. A centralized control system may prioritize received requests from the

governor(s) for operational changes, such as but not limited to, by prioritizing the largest reductions to illumination levels.

Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

BRIEF DESCRIPTION OF THE DRAWINGS

In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

FIG. 1 is a simplified plan view of an exemplary unit with automated power governing features and related system components;

FIG. 2 is a simplified perspective view of the unit of FIG. 1 in an exemplary environmental operation context;

FIG. 3 is a flow chart with exemplary logic for operating the unit of FIG. 1, such as when operating in the various environmental operation conditions illustrated by FIG. 2; and

FIG. 4 is a flow chart with other exemplary logic for operating the unit of FIG. 1, such as when operating in the various environmental operation conditions illustrated by FIG. 2 and/or in conjunction with the logic of FIG. 3.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configuration and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

FIG. 1 illustrates an exemplary outdoor display unit 10 (hereinafter “unit” for brevity). The unit 10 may include one or more electronic displays 12 (hereinafter also “display” for brevity). Each electronic display 12 may include liquid crystal displays (LCDs), organic light emitting diode displays (OLED), light emitting diode (LED) displays, plasma displays, cathode ray tube displays, rear projection displays, combinations thereof, or the like. Each electronic display 12 may include one or more backlights, such as direct backlights, edge lighting, combinations thereof, or the like. Each display 12 may not require a separate backlight, such as in the case of OLEDs, which are self-illuminating.

The units 10 may include one or more thermal management systems 20. The thermal management systems 20 may comprise one or more fans, open loop airflow pathways, closed loop airflow pathways, thermoelectric modules, air conditioning units, sensors 24, combinations thereof, or the like. The thermal management systems 20 may be, for example, without limitation, as shown and/or described in one or more of: U.S. Pat. No. 9,629,287 granted Apr. 18, 2017 entitled SYSTEM FOR USING CONSTRICTED CONVECTION WITH CLOSED LOOP COOLING SYSTEM AS THE CONVECTION PLATE, U.S. Pat. No. 10,506,738 granted Dec. 10, 2019 entitled CONSTRICTED CONVECTION COOLING FOR AN ELECTRONIC DISPLAY, U.S. Pat. No. 11,540,418 granted Dec. 27, 2022 entitled ELECTRONIC DISPLAY WITH COOLING, and/or U.S. Pat. No. 11,032,923 granted Jun. 8, 2021 entitled FIELD SERVICEABLE DISPLAY ASSEMBLY, the disclosures of each of the foregoing being hereby incorporated by reference as if fully restated herein. Other types and/or kinds of thermal management systems 20 may be utilized.

The units 10 may include one or more additional electricity consuming components 22, such as but not limited to, radio transmitters/receivers (e.g., “5G” wireless equipment), cameras, touchscreens, sensors, servers, sensors 24, combinations thereof, or the like.

The units 10 may comprise one or more sensors 24. The sensors 24 may comprise temperature sensors, pressure sensors, air quality sensors, air flow sensors, location detection devices, light sensors, color sensors, combinations thereof, or the like. Any number and/or type of sensors 24 may be provided at any number of locations within the units 10.

The control subsystem 14 may be configured to adjust electronic display 12 illumination (e.g., backlight, OLED) based on sensed ambient lighting conditions, such as by way of the one or more sensors 24. In exemplary embodiments, without limitation, the control subsystem 14 may be configured to normally increase the power level of the electronic displays 12 under relatively high ambient light conditions to provide increased image visibility. In exemplary embodiments, without limitation, the control subsystem 14 may be configured to normally decrease the power level of the electronic displays 12 under relatively low ambient light conditions to save power. Such components and/or operations may be, for example, without limitation, as shown and/or described in U.S. Pat. No. 10,440,790 granted Oct. 8, 2019 entitled ELECTRONIC DISPLAY SYSTEM WITH ILLUMINATION CONTROL, the disclosures of which are hereby incorporated by reference as if fully restated herein. Other components and/or methods of control may be utilized.

The electronic displays 12, thermal management systems 20, additional electricity consuming components 22, sensors 24, combinations thereof, or the like may be in electronic communication with a control subsystem 14, which may receive data therefrom and/or provide operational commands to such components.

The control subsystem 14 may, additionally or alternatively, be in electronic communication with a power subsystem 16. The power subsystem 16 may comprise one or more power supplies (e.g., DC power supplies), power transformers, power regulation components, capacitors, bulk energy storage devices (e.g., batteries), power converters (e.g., AC/DC converters), switches, combinations thereof, or the like. The power subsystem 16 may be electrically interposed between some or all electricity consuming components of the unit 10 (e.g., electronic displays

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12, thermal management systems 20, additional electricity consuming components 22, sensors 24, combinations thereof, or the like) and a power source 26 (e.g., external utility power supply).

Alternatively, or additionally, the control subsystem 14 may be electrically interposed between some or all electricity consuming components of the unit 10 (e.g., electronic displays 12, thermal management systems 20, additional electricity consuming components 22, sensors 24, combinations thereof, or the like) and the power source 26. The power source 26 may comprise one or more external utility power supplies, such as power generation facilities (e.g., power plants), utility lines, and/or power grids, bulk energy storage devices (e.g., batteries), local power supplies (e.g., wind power, portable or fixed power generators, solar power, combinations thereof, or the like), combinations thereof, or the like. One or more power sources 26 may be available (e.g., solar power with backup utility power). The power sources 26 may be external to the units 10. In at least the case of solar power and/or bulk energy storage devices, some or all of the power sources 26, or components thereof, may be, in whole or in part, internal to the unit 10.

Alternatively, or additionally, a power overload prevention subsystem 18 may be electrically interposed between some or all electricity consuming components of the unit 10 (e.g., electronic displays 12, thermal management systems 20, additional electricity consuming components 22, sensors 24, combinations thereof, or the like) and the power source 26. The power overload prevention subsystem 18 may comprise one or more circuit breakers, fuses, combinations thereof, or the like.

The control subsystem 14 may comprise, or be in electronic communication with (such as, by way of non-limiting example, provided at the power subsystem 16 or otherwise) one or more of: an AC power governor 28, DC power governor 30, and thermal governor 32 (collectively the “governors” for brevity). One or more of the governors may be independent components, part of the control subsystem 14, part of the power subsystem 16, combinations thereof, or the like. One or more of the governors may comprise hardware components, such as but not limited to, processors, electronic storage devices, computing devices, switches, power transformers, power limiters, power regulators, combinations thereof, or the like, and/or software components, such as but not limited to, software code, variables, algorithms, operational command subroutines, combinations thereof, or the like.

The control subsystem 14, power subsystem 16, and one or more of the governors may comprise one or more power meters and/or simulated power meters, such as shown and/or described in one or more of, and/or use one or more of the techniques shown and/or described in: U.S. Pat. No. 11,022,635 granted Jun. 1, 2021 entitled MEASURING POWER CONSUMPTION OF AN ELECTRONIC DISPLAY ASSEMBLY. Such power meters and/or simulated power meters and/or related techniques may serve as sensor(s) 24, though such is not required.

The control subsystem 14 may comprise, or be in electronic communication with, one or more network communication devices, such as for wired and/or wireless transmission and/or receipt of data. Such data may include data regarding unit 10 operation, commands, protocols, software, and/or thresholds, combinations thereof, or the like. The network communication devices may be configured to facilitate electronic communication by way of one or more internets, intranets, cellular networks, combinations thereof, or the like.

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The units 10 may comprise one or more internal support frameworks, external housings, cover panels, ventilation systems, filters, openings, combinations thereof, or the like, and may be provided in a variety of sizes, shapes, and/or configurations. Some or all of the components of the units 10 shown and/or described may be internal to the units 10 and/or external thereto.

As illustrated with particular regard to FIG. 2, a large variety of ambient weather and other operating conditions experienced by the units 10 may result in drastic fluctuations to unit 10 needs and/or operational characteristics, such as thermal management subsystem 20 demands, experienced temperatures (e.g., due to ambient temperatures, solar loading, solar angle, seasonal changes, weather patterns, combinations thereof, or the like), ambient lighting conditions (e.g., sun and moonlight movement, shade, cloud cover, precipitation, seasonal changes, combinations thereof, and the like) and resulting need to alter electronic display 12 illumination levels to render image sufficiently visible, combinations thereof, or the like. By way of non-limiting example, relatively warm temperatures and/or high solar loading may increase demand on the thermal management subsystem 20 for cooling and/or desire for increased illumination levels of the electronic displays 12 for image visibility, resulting in relatively high electrical demand on the unit 10 and/or high internal temperature levels, thereby pushing the unit 10 to the extremes of its operating abilities. Changing global weather patterns and extreme weather events may contribute to such extreme conditions. The automated power governing features shown and/or described herein may assist with managing the unit 10 such that the most extreme of such conditions, reducing the need to otherwise significantly engineer the units 10 for such unlikely or rare events. This may reduce design complexity and unit 10 cost while improving reliability and user experience. The illustrated and/or described examples of operating conditions are merely exemplary and not intended to be limiting.

FIG. 3 provides exemplary logic for operating the unit 10, such as during the various exemplary operating conditions of FIG. 2. The control subsystem 14 may operate the unit 10 normally. Such normal operations may include, for example, without limitation, displaying content at the electronic displays 12, adjusting illumination of the electronic displays 12, such as in response to ambient light readings taken by one or more of the sensors 24, operating the thermal management subsystem 20, such as by running fans at various speeds based on temperature readings taken by one or more of the sensors 24, operating customer or peripheral equipment, such as providing wayfinding information, telephonic, voice, and/or video calls, operating wireless “hot-spots”, receiving and/or processing information from the additional electronic equipment 22, such as cameras, combinations thereof, or the like. These are merely exemplary and not intended to be limiting.

While undertaking normal operations, the control subsystem 14 may monitor one or more operating conditions of the unit 10, such as by way of the sensors 24, thermal management subsystem 20, and/or power subsystem 16. In exemplary embodiments, without limitation, the AC governors 28 may periodically, continuously, randomly, combinations thereof, or the like, monitor AC current draw; the DC governor 30 may periodically, continuously, randomly, combinations thereof, or the like, monitor DC current draw; the thermal governor 32 may periodically, continuously, randomly, combinations thereof, or the like, monitor temperatures (e.g., by way of one or more sensors 24) and thermal

management subsystem **20** operations, respectively. Each unit **10** may comprise one, some, or all of the governors **28**, **30**, **32** in a same or different combination.

Where the AC governor **28** determines that AC current draw is above a predetermined threshold, the AC governor **28** may initiate a current draw mitigation mode. The predetermined threshold may be between 70-99% of the service rated expected installed AC power, though any threshold may be utilized. The current draw mitigation mode may comprise initiating a subroutine which comprises issuing commands, such as by way of the control subsystem **14**, to the electronic displays **12** to rapidly, such as in a matter of seconds or less than 1 second, begin reducing illumination levels of the electronic displays **12**, such as by dimming the backlight. This may reduce or prevent nuisance tripping of circuit breakers, such as at the power overload prevention subsystem **18**. In this way, the current draw mitigation mode may replace and/or override the normal operations. The unit **10** may remain in the current draw mitigation mode until AC current draw is below the predetermined threshold, such as by at least a margin to prevent continued movement between normal operation mode and current draw mitigation mode. Where the AC governor **28** determines that AC current draw is below the predetermined threshold, the AC governor **28** continue with normal operations.

The AC governor **28**, control subsystem **14**, and/or power subsystem **16** may monitor current supplied, such as on a continual, periodic, and/or random basis. Where the current draw is below an expected level, an electronic notification may be automatically generated and transmitted, such as by way of one or more network communication devices, to one or more remote electronic devices (e.g., computers, smart phones, tablets, servers, etc.). The network communication devices and transmission may be made by way of one or more internets, intranets, cellular networks, combinations thereof, or the like.

Expected power supply levels, including current supply levels, may vary based on unit **10** configuration, such as size, number, and/or type of the electronic displays **12** installed, anticipated driving levels for the electronic displays **12**, other equipment **22** installed, expected ambient conditions, combinations thereof, and the like. Such power consumption levels may vary from approximately 200 watts to 5000+ watts, though any power level may be expected. Such expected current may vary from less than 2 amps to over 25 amps, though any current level may be expected. Circuit breakers may be configured to trip at less than 5 amps to over 30 amps, though any threshold may be utilized. These are provided by way of non-limiting example.

Actual power supplied may vary based on power source **26** type and/or operational fluctuations, economic issues (e.g., operator unable or unwilling to pay for certain power supply at peak times, by way of non-limiting example), combinations thereof, or the like.

Where the DC governor **30** determines that DC current demand is above a predetermined threshold, the DC governor **30** may initiate a power demand mitigation mode. The predetermined threshold may be reflective of limits of one or more installed power supplies, such as forming part of the power subsystem **16**. The predetermined threshold may be between 80-100% of the limits of the one or more installed power supplies, though any amount may be utilized. The power demand mitigation mode may comprise initiating a subroutine which comprises issuing commands, such as by way of the control subsystem **14**, to the electronic displays **12** to at a relatively moderate pace, such as in a matter of seconds (e.g., between 3-60 seconds) or minutes, begin

reducing illumination levels of the electronic displays **12**, such as by dimming the backlight. This may reduce or prevent the one or more power supplies from reaching their maximum limit and turning off. In this way, the power demand mitigation mode may replace and/or override the normal operations. The unit **10** may remain in the power demand mitigation mode until DC current demand is below the predetermined threshold, such as by at least a margin to prevent continued movement between normal operation mode and power demand mitigation mode. Where the DC governor **30** determines that DC current demand is below the predetermined threshold, the DC governor **30** may continue with normal operations.

If the one or more power supplies reach their current limit, they may be configured to automatically shut off and be restarted.

Where the thermal governor **32** determines that the thermal management subsystem **20** is operating at a predetermined capacity threshold and internal temperatures, such as determined by the one or more sensors **24**, are above a predetermined temperature threshold, the thermal governor **32** may initiate a temperature rise mitigation mode. Stated another way, the thermal management subsystem **20** may be performing at a maximum level for heat removal or some threshold thereof. For example, without limitation, the predetermined capacity threshold may be between 80-100% of the capacity of the thermal management subsystem **20** (e.g., fans operating at 80-100% of maximum speed), though any amount may be utilized. The predetermined temperature threshold may be the same or different for each sensor **24**, such as based on location and/or tolerance of local components. In exemplary embodiments, without limitation, the internal temperature condition for entering the temperature rise mitigation mode may be met where any one of the internal temperatures are above the predetermined temperature threshold and/or the respective predetermined temperature threshold for the sensor **24**. In other exemplary embodiments, without limitation, a plurality, or all, of the internal temperatures are above the predetermined temperature threshold and/or the respective predetermined temperature threshold for the sensor **24** before the temperature conditions for entering the temperature rise mitigation mode are met.

The temperature rise mitigation mode may comprise initiating a subroutine which comprises issuing commands, such as by way of the control subsystem **14**, to the electronic displays **12** at a relatively slow pace, such as in a matter of minutes (e.g., between 5-60 minutes), begin reducing illumination levels of the electronic displays **12**, such as by dimming the backlight, such as down to a zero level. This may reduce or prevent damage to temperature sensitive components of the unit **10**. In this way, the temperature rise mitigation mode may replace and/or override the normal operations. The unit **10** may remain in the temperature rise mitigation mode until the internal temperatures and/or thermal management subsystem **20** capacity is below the respective predetermined thresholds, such as by at least a margin to prevent continued movement between normal operation mode and temperature rise mitigation mode. Where the thermal governor **32** determines that internal temperatures and/or thermal management subsystem **20** capacity are below the predetermined threshold, the thermal governor **32** may continue with normal operations.

These governors **28**, **30**, and **32** may be particularly important for preventing shut down of the units **10** and/or related components, such as the thermal management subsystems **20** which, if shut down, particularly in a moment of existing extreme operating conditions, may trigger a rapid

rise in internal temperatures and comprise of the units 10. Thus, the governors 28, 30, and 32 and related operations may serve to maintain operations of the unit 10, such as the thermal management subsystem 20, and/or minimize disruption to user experiences.

The various predetermined thresholds and/or criteria for the governors 28, 30, and/or 32 may be set and/or varied by programming, such as by way of receipt of authenticated change commands from one or more remote electronic devices.

The governors 28, 30, and/or 32 may operate independently from one another, in exemplary embodiments without limitation. The analysis undertaken by each of the governors 28, 30, and/or 32 may be performed in parallel and/or in any sequence.

The integrated power governing features may alternatively, or additionally, permit the reduction in power supplies, such as of the power subsystem 16, for a unit 10 and/or reduce power consumption for a unit 10, among other benefits. This may allow provided power supplies, such as of the power subsystem 16, to operate closer to maximum capacity, whereby increased efficiencies are generally found, thereby increasing operational efficiency of the unit 10 and reducing costs of manufacture, among other benefits.

As illustrated with particular regard to FIG. 4, requests for operational changes may be prioritized and/or centralized for disposition. For example, without limitation, operational change requests made by the governors 28, 30, and/or 32 may be centrally dispositioned, such as at the control subsystem 14. Alternatively, or additionally, the governors 28, 30, and/or 32 may be in intercommunication and/or communication with a particular one of the governors 28, 30, and/or 32 which may be designated for arbitrating and/or prioritizing incoming requests. Where multiple requests for operational changes are received, such as from the different governors 28, 30, and/or 32 at the control subsystem 14, the control subsystem 14 or other dispositioning unit (e.g., designated one of the governors 28, 30, and/or 32) may be configured to prioritize the requests based on an amount of operational adjustment requested. For example, the largest operational change may be prioritized over the other requests. In exemplary embodiments, this may require acting on the request which results in the largest power and/or electronic display 12 illumination level reduction.

The remaining requests may be discarded, ignored, and/or placed on hold. The remaining requests may be acted on subsequently, such as when the particular request being acted on is no longer valid, and/or the requests may be discarded as the various requests become no longer valid. For example, without limitation, the highest prioritized request may comprise the largest illumination level reduction which may be sufficient to request the lower-level requests. As another example, without limitation, once the highest prioritized request is performed for a period of time, the request may drop off due to sufficient changes to operational conditions that render the request no longer needed, and the control subsystem 14, for example, may move to a secondary request, tertiary request, etc. which remain valid based on updated operating conditions.

While three specific governors 28, 30, and/or 32 may be utilized in exemplary embodiments, one or more additional governors 31 may optionally be utilized in place of, or in addition to, the governors 28, 30, and/or 32. For example, without limitation, one of the additional governors 31 may be configured to trigger electronic display 12 dimming and/or other power consumption reduction efforts where one or more operational failures in the thermal management

system 20 are detected. For example, without limitation, where one or more fan failures are detected. Such operational failures may be detected through lack of signal response, lack of power supply, combinations thereof, or the like. In other exemplary embodiments, without limitation, one of the additional governors 31 may be configured to trigger electronic display 12 dimming and/or other power consumption reduction efforts where external power supply changes or ceases, such as from one or more solar panels, utility power supplies, wind turbines, combinations thereof, or the like.

Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention.

Certain operations described herein may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, combinations thereof, and the like, configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing devices. The electronic devices may comprise personal computers, smartphones, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein. The electronic devices, including but not necessarily limited to the electronic storage devices, databases, controllers, or the like, may comprise and/or be configured to hold solely non-transitory signals.

What is claimed is:

1. A display unit for prioritizing requested operational updates, said display unit comprising:
 - an electronic display;
 - a thermal management subsystem;
 - one or more sensors; and
 - a control subsystem in electronic communication with the electronic display, the one or more sensors, and the thermal management subsystem, where the control subsystem comprises one or more non-transitory electronic storage devices with software instructions, which when executed, configure one or more processors to:
 - operate the display unit based on operational parameters which are specific to readings from the one or more sensors;
 - receive requests to update the operational parameters for the display unit;
 - for each of the requests, determine a change in value to an associated one or ones of the operational parameters or an expected change in energy consumption for the display unit if the respective request were implemented; and

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prioritize the requests for implementation based on the change in value or the expected change in energy consumption.

2. The display unit of claim 1 wherein:

the software instructions, when executed, configure the one or more processors to: for each of the requests, determine the change in value to the associated one or ones of the operational parameters or the expected change in energy consumption for the display unit if the respective request were implemented on an absolute basis.

3. The display unit of claim 1 wherein:

the software instructions, when executed, configure the one or more processors to: implement the requests in accordance with the prioritization.

4. The display unit of claim 1 wherein:

the software instructions, when executed, configure the one or more processors to: discard or ignore all of the requests aside from a highest prioritized one of the requests.

5. The display unit of claim 1 wherein:

the operational parameters in the requests comprise illumination setpoints for the electronic display.

6. The display unit of claim 5 wherein:

the one or more sensors comprise at least one ambient light sensor; and

the illumination setpoints are specific to ambient light readings from the at least one ambient light sensor.

7. The display unit of claim 5 wherein:

the one or more sensors comprise at least one temperature light sensor; and

the illumination setpoints are specific to temperature readings from the at least one temperature sensor.

8. The display unit of claim 5 wherein:

the electronic display comprises a directly backlit liquid crystal display (LCD).

9. The display unit of claim 1 wherein:

the operational parameters in the requests comprise setpoints for the thermal management subsystem.

10. The display unit of claim 9 wherein:

the one or more sensors comprise at least one temperature sensor; and

the setpoints are specific to temperature readings from the at least one temperature sensor.

11. The display unit of claim 10 wherein:

the thermal management subsystem comprises one or more fans; and

the setpoints comprise fan speeds levels.

12. The display unit of claim 11 wherein:

the one or more fans comprise a plurality of fans; and the setpoints are different for at least some of the fans.

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13. The display unit of claim 10 wherein:

the at least one temperature sensor comprises a plurality of temperature sensors spaced apart within the display unit; and

the setpoints are different for at least some of the temperature sensors.

14. The display unit of claim 1 wherein:

the control subsystem comprises at least one of: an alternating current (AC) governor, a direct current (DC) governor, and a thermal governor; and

the software instructions, when executed, configure the one or more processors to: assign each of the requests to a respective one of the governors for execution.

15. The display unit of claim 1 wherein:

the software instructions, when executed, configure the one or more processors to: receive readings from the one or more sensors; and operate the display unit in accordance with the readings and the operational parameters.

16. A display unit for prioritizing requested operational updates, said display unit comprising:

an electronic display;

a thermal management subsystem;

one or more sensors; and

a control subsystem in electronic communication with the electronic display, the one or more sensors, and the thermal management subsystem, where the control subsystem comprises one or more non-transitory electronic storage devices with software instructions, which when executed, configure one or more processors to:

operate the display unit based on operational parameters which are specific to readings from the one or more sensors;

receive requests to update the operational parameters for the display unit;

prioritize the requests for implementation; and

discard or ignore the requests aside from a highest prioritized one of the requests.

17. A display unit for prioritizing requested operational updates, said display unit comprising: an electronic display;

a thermal management subsystem; one or more sensors; and a control subsystem in electronic communication with the

electronic display, the one or more sensors, and the thermal management subsystem, where the control subsystem comprises one or more non-transitory electronic storage devices

with software instructions, which when executed, configure one or more processors to: operate the display unit based on

operational parameters which are specific to readings from the one or more sensors; receive requests to update the

operational parameters for the display unit, wherein the operational parameters in the requests comprise illumination

setpoints for the electronic display or setpoints for the thermal management subsystem; and prioritize the requests

for implementation by magnitude of proposed change.

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